

Foundation Mathematics 1017SCG
Week 10 Workshop Solutions

Remember that the method shown in the solutions is not the only way to solve the question. There are usually multiple ways to solve the same question.

1. (a)

$$\begin{aligned}y &= 4x - 7 \\y &= 4x^1 - 7 \\\frac{dy}{dx} &= 4 \times 1 \times x^{1-1} + 0 \\&= 4x^0 \\&= 4\end{aligned}$$

(b)

$$\begin{aligned}y &= -7x + 2 \\y &= -7x^1 + 2 \\\frac{dy}{dx} &= -7 \times 1 \times x^{1-1} + 0 \\&= -7x^0 \\&= -7\end{aligned}$$

(c)

$$\begin{aligned}y &= 3x^2 - 4x + 8 \\\frac{dy}{dx} &= 3 \times 2 \times x^{2-1} - 4 \times 1 \times x^{1-1} + 0 \\&= 6x^1 - 4x^0 \\&= 6x - 4\end{aligned}$$

(d)

$$\begin{aligned}y &= 5x^3 - 8x^2 + 9x + 1 \\\frac{dy}{dx} &= 5 \times 3 \times x^{3-1} - 8 \times 2 \times x^{2-1} + 9 \times 1 \times x^{1-1} + 0 \\&= 15x^2 - 16x + 9\end{aligned}$$

(e)

$$\begin{aligned}y &= -6x^4 + 10x^3 \\\frac{dy}{dx} &= -6 \times 4 \times x^{4-1} + 10 \times 3 \times x^{3-1} \\&= -24x^3 + 30x^2\end{aligned}$$

(f)

$$\begin{aligned}y &= 6 \sin(x) + x^2 \\\frac{dy}{dx} &= 6 \cos(x) + 2x^{2-1} \\&= 6 \cos(x) + 2x\end{aligned}$$

(g)

$$\begin{aligned}y &= 9 \cos(x) - 3 \sin(x) \\ \frac{dy}{dx} &= 9 \times -\sin(x) - 3 \cos(x) \\ &= -9 \sin(x) - 3 \cos(x)\end{aligned}$$

(h)

$$\begin{aligned}y &= -4 \cos(x) + 7 \sin(x) \\ \frac{dy}{dx} &= -4 \times -\sin(x) + 7 \cos(x) \\ &= 4 \sin(x) + 7 \cos(x)\end{aligned}$$

(i)

$$\begin{aligned}y &= 3e^x + 4 \\ \frac{dy}{dx} &= 3e^x + 0 \\ &= 3e^x\end{aligned}$$

(j)

$$\begin{aligned}y &= 6e^x + x^2 \\ \frac{dy}{dx} &= 6e^x + 2 \times x^{2-1} \\ &= 6e^x + 2x\end{aligned}$$

(k)

$$\begin{aligned}y &= -10e^x - x^3 \\ \frac{dy}{dx} &= -10e^x - 3 \times x^{3-1} \\ &= -10e^x - 3x^2\end{aligned}$$

(l)

$$\begin{aligned}y &= 2 \ln(x) \\ \frac{dy}{dx} &= 2 \times \frac{1}{x} \\ &= \frac{2}{x}\end{aligned}$$

(m)

$$\begin{aligned}y &= -4 \ln(x) \\ \frac{dy}{dx} &= -4 \times \frac{1}{x} \\ &= -\frac{4}{x}\end{aligned}$$

(n)

$$\begin{aligned}y &= -7 \ln(x) + 4 \\ \frac{dy}{dx} &= -7 \times \frac{1}{x} + 0 \\ &= -\frac{7}{x}\end{aligned}$$

2. (a)

$$\begin{aligned}y &= \frac{4}{x^2} \\ &= 4x^{-2}\end{aligned}$$

$$\begin{aligned}\frac{dy}{dx} &= 4 \times -2 \times x^{-2-1} \\ &= -8x^{-3} \\ &= -\frac{8}{x^3}\end{aligned}$$

(b)

$$\begin{aligned}y &= -3\sqrt{x} \\ &= -3 \times x^{\frac{1}{2}}\end{aligned}$$

$$\begin{aligned}\frac{dy}{dx} &= -3 \times \frac{1}{2} \times x^{\frac{1}{2}-1} \\ &= -\frac{3}{2} \times x^{-\frac{1}{2}} \\ &= -\frac{3}{2x^{\frac{1}{2}}} \\ &= -\frac{3}{2\sqrt{x}}\end{aligned}$$

(c)

$$\begin{aligned}y &= 5 \ln(x^2) \\ &= 5 \times 2 \times \ln(x) \quad (\text{using log laws}) \\ &= 10 \ln(x) \\ \frac{dy}{dx} &= 10 \times \frac{1}{x} \\ &= \frac{10}{x}\end{aligned}$$

3.

$$y = 2x^3 + 3x^2 - 36x + 12$$

(a) To find the turning points of the function, we will first need to find $\frac{dy}{dx}$.

$$\frac{dy}{dx} = 6x^2 + 6x - 36$$

To find turning points, let $\frac{dy}{dx} = 0$.

$$6x^2 + 6x - 36 = 0$$

$$6(x^2 + x - 6) = 0$$

$$6(x + 3)(x - 2) = 0$$

$$x = -3, x = 2$$

Therefore the function has two turning points which occur at $x = -3$ and $x = 2$. We will now substitute these x values into the original function to find the y -coordinate of the turning points.

$$\begin{aligned}\text{When } x = -3, y &= 2 \times (-3)^3 + 3 \times (-3)^2 - 36 \times -3 + 12 \\ &= 93\end{aligned}$$

$$\begin{aligned}\text{When } x = 2, y &= 2 \times 2^3 + 3 \times 2^2 - 36 \times 2 + 12 \\ &= -32\end{aligned}$$

Therefore the function has turning points at $(-3, 93)$ and $(2, -32)$.

Remember that when writing an ordered pair, the x -coordinate is written first.

- (b) To find the greatest and least value of the function, we need to examine turning points (found in part (a)) as well as the end points of the domain.

$$\begin{aligned}\text{When } x = -6, y &= 2 \times (-6)^3 + 3 \times (-6)^2 - 36 \times -6 + 12 \\ &= -96\end{aligned}$$

$$\begin{aligned}\text{When } x = 4, y &= 2 \times 4^3 + 3 \times 4^2 - 36 \times 4 + 12 \\ &= 44\end{aligned}$$

Therefore the least value of the function is -96 which occurs at $x = -6$. The greatest value of the function is 93 which occurs at $x = -3$.

4.

$$y = 3e^x - 12x$$

- (a) To find the turning points of the function, we will first need to find $\frac{dy}{dx}$.

$$\frac{dy}{dx} = 3e^x - 12$$

To find turning points, let $\frac{dy}{dx} = 0$.

$$3e^x - 12 = 0$$

$$3e^x = 12$$

$$e^x = 4$$

$$\ln(e^x) = \ln(4)$$

$$x \ln(e) = \ln(4)$$

$$x = \ln(4)$$

Therefore the function has one turning point which occurs at $x = \ln(4)$. We will now substitute this x value into the original function to find the y -coordinate of the turning point.

$$\begin{aligned}\text{When } x = \ln(4), y &= 3e^{\ln(4)} - 12 \times \ln(4) \\ &= -4.64\end{aligned}$$

Therefore the function has a turning point at (1.38,-4.64).

Remember that when writing an ordered pair, the x-coordinate is written first.

- (b) To find the greatest and least value of the function, we need to examine the turning point (found in part (a)) as well as the end points of the domain.

$$\begin{aligned}\text{When } x = 0, y &= 3e^0 - 12 \times 0 \\ &= 3\end{aligned}$$

$$\begin{aligned}\text{When } x = 2, y &= 3e^2 - 12 \times 2 \\ &= -1.83\end{aligned}$$

Therefore the least value of the function is -4.64 which occurs at $x = \ln(4) = 1.38$.
The greatest value of the function is 3 which occurs at $x = 0$.

5. We have been asked to find the tangent line for the function $y = x^2 - 4x + 4$ at $x = 3$.
Remember that a tangent line will have the following properties:

- It is a straight line [which can be found using $y - y_1 = m(x - x_1)$].
- The tangent line passes through the **same point** as the original function at $x = 3$.
- The tangent line will have the **same gradient** as the original function at $x = 3$.

$$\begin{aligned}\text{When } x = 3, y &= 3^2 - 4 \times 3 + 4 \\ &= 1\end{aligned}$$

Therefore, the tangent line will pass through the point (3,1).

To find the gradient of the tangent line, we will need to find $\frac{dy}{dx}$.

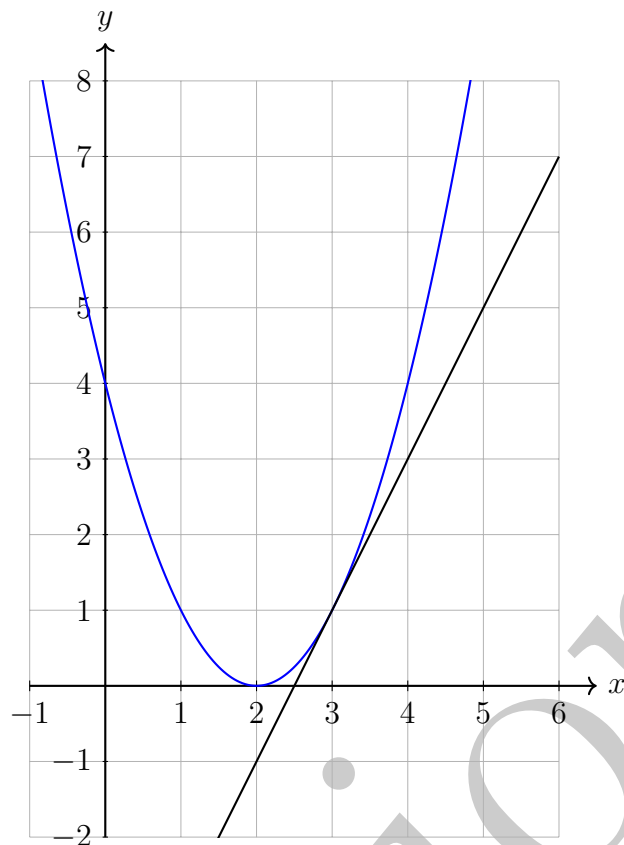
$$\begin{aligned}\frac{dy}{dx} &= 2x - 4 \\ \text{When } x = 3, \frac{dy}{dx} &= 2 \times 3 - 4 \\ &= 2\end{aligned}$$

Therefore, the gradient of the tangent line is 2.

$$\begin{aligned}y - y_1 &= m(x - x_1) \\ y - 1 &= 2(x - 3) \\ y - 1 &= 2x - 6 \\ y &= 2x - 5\end{aligned}$$

Therefore the tangent to the function $y = x^2 - 4x + 4$ at $x = 3$ is $y = 2x - 5$.

While not asked in the question, below is a graph of the original function in blue and the tangent function in black. Note that at $x = 3$, the tangent line passes through the same point as the original function and matches the gradient of the original function.



6. We have been asked to find the tangent line for the function $y = -x^2 - x + 6$ at $x = 1$. Remember that a tangent line will have the following properties:
- It is a straight line [which can be found using $y - y_1 = m(x - x_1)$.]
 - The tangent line passes through the **same point** as the original function at $x = 1$.
 - The tangent line will have the **same gradient** as the original function at $x = 1$.

$$\begin{aligned}\text{When } x = 1, \quad y &= -1 \times 1^2 - 1 + 6 \\ &= 4\end{aligned}$$

Therefore, the tangent line will pass through the point (1,4).

To find the gradient of the tangent line, we will need to find $\frac{dy}{dx}$.

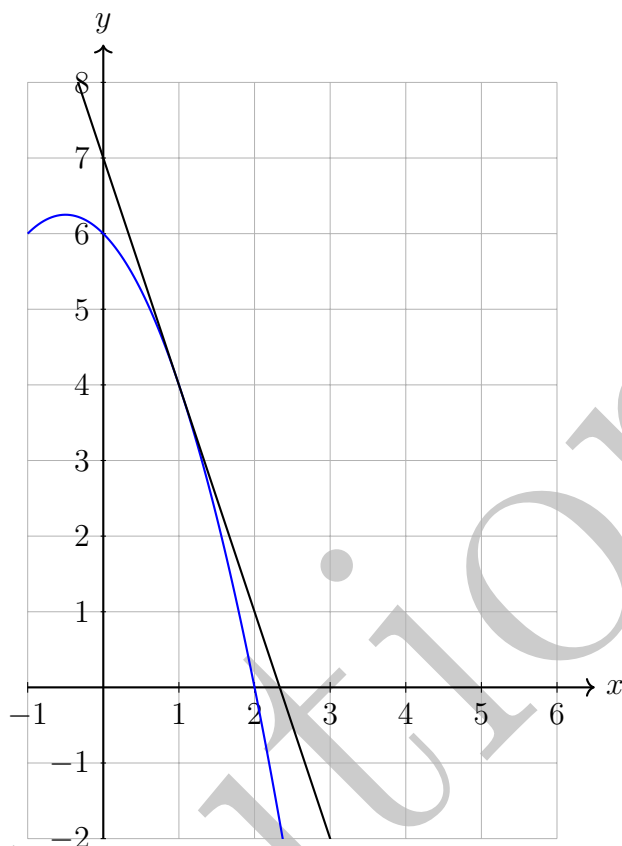
$$\begin{aligned}\frac{dy}{dx} &= -2x - 1 \\ \text{When } x = 1, \quad \frac{dy}{dx} &= -2 \times 1 - 1 \\ &= -3\end{aligned}$$

Therefore, the gradient of the tangent line is -3.

$$\begin{aligned}y - y_1 &= m(x - x_1) \\ y - 4 &= -3(x - 1) \\ y - 4 &= -3x + 3 \\ y &= -3x + 7\end{aligned}$$

Therefore the tangent to the function $y = -x^2 - x + 6$ at $x = 1$ is $y = -3x + 7$.

While not asked in the question, below is a graph of the original function in blue and the tangent function in black. Note that at $x = 3$, the tangent line passes through the same point as the original function and matches the gradient of the original function.



7. We have been asked to find the tangent line for the function $y = 4\sin(x) - 2$ at $x = \pi$. Remember that a tangent line will have the following properties:

- It is a straight line [which can be found using $y - y_1 = m(x - x_1)$].
- The tangent line passes through the **same point** as the original function at $x = \pi$.
- The tangent line will have the **same gradient** as the original function at $x = \pi$.

$$\begin{aligned}\text{When } x = \pi, y &= 4\sin(\pi) - 2 \\ &= -2\end{aligned}$$

Therefore, the tangent line will pass through the point $(\pi, -2)$. Alternatively, we can round this to $(3.14, -2)$.

To find the gradient of the tangent line, we will need to find $\frac{dy}{dx}$.

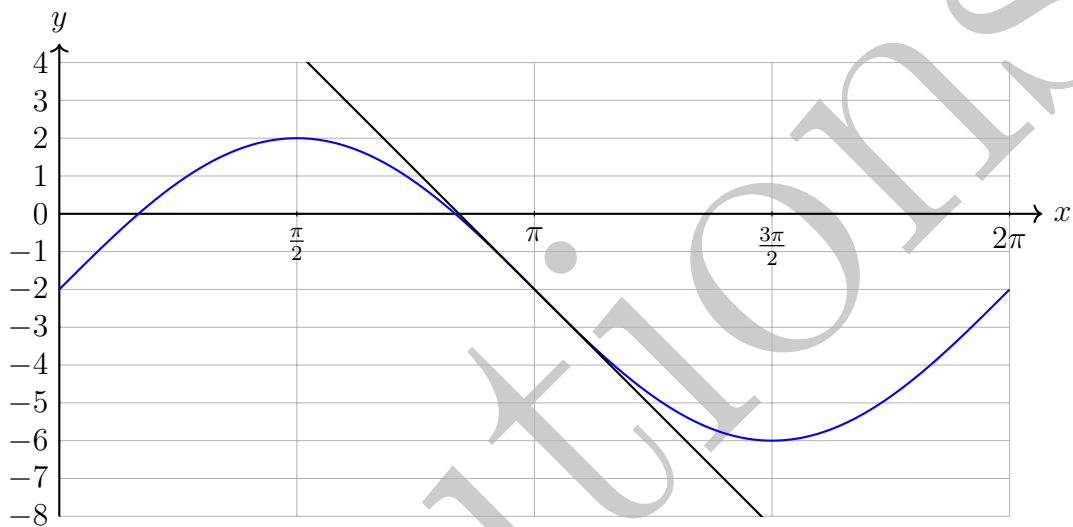
$$\begin{aligned}\frac{dy}{dx} &= 4\cos(x) \\ \text{When } x = \pi, \frac{dy}{dx} &= 4 \times \cos(\pi) \\ &= -4\end{aligned}$$

Therefore, the gradient of the tangent line is -4.

$$\begin{aligned}y - y_1 &= m(x - x_1) \\y + 2 &= -4(x - \pi) \\y + 2 &= -4x + 4\pi \\y &= -4x + 4\pi - 2 \\y &= -4x + 10.57\end{aligned}$$

Therefore the tangent to the function $y = 4 \sin(x) - 2$ at $x = 2$ is $y = -4x + 10.57$.

While not asked in the question, below is a graph of the original function in blue and the tangent function in black. Note that at $x = \pi$, the tangent line passes through the same point as the original function and matches the gradient of the original function.



8.

$$G(t) = 20t(9 - t^2) + 450$$

(a) The initial population is when $t = 0$.

$$G(0) = 20 \times 0 \times (9 - 0^2) + 450 = 450$$

There was initially 450 seagulls.

(b)

$$\begin{aligned}G(t) &= 20t(9 - t^2) + 450 \\&= 180t - 20t^3 + 450 \\\frac{dG}{dt} &= 180 - 60t^2\end{aligned}$$

$$\text{When } t = 1, \frac{dG}{dt} = 180 - 60 \times 1^2 = 120$$

Therefore after one month, the seagull population is growing at a rate of 120 seagulls/month.

(c)

$$G(3) = 20 \times 3(9 - 3^2) + 450 = 450$$

There were 450 seagulls at the end of the research project.

- (d) The greatest population of seagulls can either occur at an end point ($t = 0$ or $t = 3$) or at a turning point. To find turning points, let $\frac{dG}{dt} = 0$.

$$180 - 60t^2 = 0$$

$$60t^2 = 180$$

$$t^2 = 3$$

$$t = \pm\sqrt{3}$$

We only need to test $t = \sqrt{3}$ as $t = -\sqrt{3}$ falls outside the domain of $G(t)$ ($0 \leq t \leq 3$).

$$G(\sqrt{3}) = 20 \times \sqrt{3}(9 - (\sqrt{3})^2) + 450 \approx 658$$

Therefore, the greatest population of seagulls was approximately 658.

9.

$$A = 2e^t - 5t + 4$$

- (a) When the supervisor arrived on the oil rig, $t = 0$. To find the area of the oil spill when the supervisor arrived at the oil rig, we need to substitute $t = 0$ into the original equation.

$$\begin{aligned} \text{When } t = 0, A &= 2e^0 - 5 \times 0 + 4 \\ &= 6 \end{aligned}$$

Therefore, when the supervisor arrived at the oil rig, the area of the oil patch was 6km^2 .

- (b) To find the area of the oil spill two hours after the supervisor arrived at the oil rig, we need to substitute $t = 2$ into the original equation.

$$\begin{aligned} \text{When } t = 2, A &= 2e^2 - 5 \times 2 + 4 \\ &= 8.78 \end{aligned}$$

Therefore, two hours after the supervisor arrived at the oil rig, the area of the oil patch was 8.78km^2 .

- (c) To find the rate of change of the area of the oil patch, we need to find $\frac{dA}{dt}$.

$$\frac{dA}{dt} = 2e^t - 5$$

To find the rate of change of the area of the oil patch two hours after the supervisor arrived at the oil rig, we need to substitute $t = 2$ into $\frac{dA}{dt}$.

$$\begin{aligned} \text{When } t = 2, \frac{dA}{dt} &= 2e^2 - 5 \\ &= 9.78 \end{aligned}$$

Therefore, two hours after the supervisor arrived at the oil rig, the area of the oil patch was increasing at a rate of 9.78km^2 per hour.

- (d)
- If $\frac{dA}{dt} > 0$, the area of the oil patch is increasing.
 - If $\frac{dA}{dt} = 0$, the area of the oil patch is remaining constant. It is important to understand that this does not mean the area of the oil patch is zero. The area of the oil patch may be quite large, but it is not increasing nor decreasing in size.

- If $\frac{dA}{dt} < 0$, the area of the oil patch is decreasing.

10. We are given the the function $y = -x^2 + 6x$.

$$\begin{aligned}\frac{dy}{dx} &= -1 \times 2 \times x^{2-1} + 6 \times 1 \times x^{1-1} \\ &= -2x + 6\end{aligned}$$

(a) We have been asked to find the equation of the tangent line when $x = 2$.

$$\text{When } x = 2, y = -2^2 + 6 \times 2 = 8$$

$$\text{When } x = 2, \frac{dy}{dx} = 2$$

Therefore the tangent line has a gradient of 2 and passes through the point (2,8).

$$y - y_1 = m(x - x_1)$$

$$y - 8 = 2(x - 2)$$

$$y - 8 = 2x - 4$$

$$y = 2x + 4$$

Therefore the tangent line to the curve $y = -x^2 + 6x$ at $x = 2$ is $y = 2x + 4$.

(b) We have been asked to find the equation of the tangent line when $y = 5$. The first step is to find the corresponding x values.

$$5 = -x^2 + 6x$$

$$x^2 - 6x + 5 = 0$$

$$(x - 1)(x - 5) = 0$$

$$x = 1 \text{ or } x = 5$$

We will find the tangent line for $x = 1$ first.

$$\text{When } x = 1, y = -1^2 + 6 \times 1 = 5$$

$$\text{When } x = 1, \frac{dy}{dx} = 4$$

Therefore the tangent line at $x = 1$ has a gradient of 4 and passes through the point (1,5).

$$y - y_1 = m(x - x_1)$$

$$y - 5 = 4(x - 1)$$

$$y - 5 = 4x - 4$$

$$y = 4x + 1$$

Now we will find the tangent line for $x = 5$.

$$\text{When } x = 5, y = -5^2 + 6 \times 5 = 5$$

$$\text{When } x = 5, \frac{dy}{dx} = -4$$

Therefore the tangest line at $x = 5$ has a gradient of -4 and passes through the point $(5,5)$.

$$\begin{aligned}y - y_1 &= m(x - x_1) \\y - 5 &= -4(x - 5) \\y - 5 &= -4x + 20 \\y &= -4x + 25\end{aligned}$$

Therefore, the tangent lines when $y = 5$ are $y = 4x + 1$ and $y = -4x + 25$.

- (c) We have been asked to find the equation of the tangent line when the curve cuts the x -axis.

$$\begin{aligned}-x^2 + 6x &= 0 \\x(-x + 6) &= 0 \\x = 0 \text{ or } x &= 6\end{aligned}$$

We will find the tangent line for $x = 0$ first.

$$\text{When } x = 0, y = -0^2 + 6 \times 0 = 0$$

$$\text{When } x = 0, \frac{dy}{dx} = 6$$

Therefore the tangest line at $x = 0$ has a gradient of 6 and passes through the point $(0,0)$.

$$\begin{aligned}y - y_1 &= m(x - x_1) \\y - 0 &= 6(x - 0) \\y &= 6x\end{aligned}$$

Now we will find the tangent line for $x = 6$.

$$\text{When } x = 6, y = -6^2 + 6 \times 6 = 0$$

$$\text{When } x = 6, \frac{dy}{dx} = -6$$

Therefore the tangest line at $x = 6$ has a gradient of -6 and passes through the point $(6,0)$.

$$\begin{aligned}y - y_1 &= m(x - x_1) \\y - 0 &= -6(x - 6) \\y &= -6x + 36\end{aligned}$$

Therefore, the tangent lines where the curve cuts the x -axis are $y = 6x$ and $y = -6x + 36$.

11.

$$\begin{aligned}y &= x^2 + 5x - 6 \\\frac{dy}{dx} &= 2x + 5\end{aligned}$$

To begin, we need to find the x values of the tangent lines. When the curve cuts the x -axis, $y = 0$.

$$\begin{aligned}x^2 + 5x - 6 &= 0 \\(x + 6)(x - 1) &= 0 \\x &= -6 \text{ or } x = 1\end{aligned}$$

We will start by finding the tangent line at $x = -6$.

$$\text{When } x = -6, \frac{dy}{dx} = -7$$

Therefore, the tangent line at $x = -6$ has a gradient of -7 and passes through the point $(-6, 0)$

$$\begin{aligned}y - y_1 &= m(x - x_1) \\y - 0 &= -7(x - -6) \\y &= -7(x + 6) \\y &= -7x - 42\end{aligned}$$

We will now find the tangent line at $x = 1$.

$$\text{When } x = 1, \frac{dy}{dx} = 7$$

Therefore, the tangent line at $x = 1$ has a gradient of 7 and passes through the point $(1, 0)$

$$\begin{aligned}y - y_1 &= m(x - x_1) \\y - 0 &= 7(x - 1) \\y &= 7x - 7\end{aligned}$$

Therefore, the equations of the tangents when the curve cuts the x -axis are

$$\begin{aligned}y &= -7x - 42 \\y &= 7x - 7\end{aligned}$$

To find where the two tangents intersect, we need to solve the simultaneous equations above. This gives

$$\begin{aligned}-7x - 42 &= 7x - 7 \\14x &= -35 \\x &= -2.5\end{aligned}$$

Substituting $x = -2.5$ into the first equation gives

$$y = -7 \times -2.5 - 42 = -24.5$$

Therefore, the two tangent lines intersect at $x = -2.5, y = -24.5$.

12. (a) Let $y = f(x) = 2x - 5$
Start by calculating $f(x + h)$

$$\begin{aligned}f(x + h) &= 2(x + h) - 5 \\&= 2x + 2h - 5\end{aligned}$$

$$\begin{aligned}
\frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h \rightarrow 0} \frac{(2x+2h-5) - (2x-5)}{h} \\
&= \lim_{h \rightarrow 0} \frac{2x+2h-5-2x+5}{h} \\
&= \lim_{h \rightarrow 0} \frac{2h}{h} \\
&= \lim_{h \rightarrow 0} 2 \\
&= 2
\end{aligned}$$

(b) Let $y = f(x) = 5x^2$

Start by calculating $f(x+h)$

$$\begin{aligned}
f(x+h) &= 5(x+h)^2 \\
&= 5(x^2 + 2xh + h^2) \\
&= 5x^2 + 10xh + 5h^2
\end{aligned}$$

$$\begin{aligned}
\frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h \rightarrow 0} \frac{5x^2 + 10xh + 5h^2 - 5x^2}{h} \\
&= \lim_{h \rightarrow 0} \frac{h(10x + 5h)}{h} \\
&= \lim_{h \rightarrow 0} 10x + 5h \\
&= 10x
\end{aligned}$$

(c) Let $y = f(x) = 3x^2 - 4x$

Start by calculating $f(x+h)$

$$\begin{aligned}
f(x+h) &= 3(x+h)^2 - 4(x+h) \\
&= 3(x^2 + 2xh + h^2) - 4(x+h) \\
&= 3x^2 + 6xh + 3h^2 - 4x - 4h
\end{aligned}$$

$$\begin{aligned}
\frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h \rightarrow 0} \frac{(3x^2 + 6xh + 3h^2 - 4x - 4h) - (3x^2 - 4x)}{h} \\
&= \lim_{h \rightarrow 0} \frac{6xh + 3h^2 - 4h}{h} \\
&= \lim_{h \rightarrow 0} \frac{h(6x + 3h - 4)}{h} \\
&= \lim_{h \rightarrow 0} 6x + 3h - 4 \\
&= 6x - 4
\end{aligned}$$

(d) Let $y = f(x) = x^2 - 2x + 7$

Start by calculating $f(x+h)$

$$\begin{aligned}
f(x+h) &= (x+h)^2 - 2(x+h) + 7 \\
&= x^2 + 2xh + h^2 - 2x - 2h + 7
\end{aligned}$$

$$\begin{aligned}
\frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
&= \lim_{h \rightarrow 0} \frac{(x^2 + 2xh + h^2 - 2x - 2h + 7) - (x^2 - 2x + 7)}{h} \\
&= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 2x - 2h + 7 - x^2 + 2x - 7}{h} \\
&= \lim_{h \rightarrow 0} \frac{2xh + h^2 - 2h}{h} \\
&= \lim_{h \rightarrow 0} \frac{h(2x + h - 2)}{h} \\
&= \lim_{h \rightarrow 0} 2x + h - 2 \\
&= 2x - 2
\end{aligned}$$